DP Computer Science: A 2.4.4 Encryption and Digital Certificates

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1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

- Differentiate between **symmetric** and **asymmetric** cryptography
- Explain the role of **digital certificates** in establishing secure network connections
- Describe the use of **public** and **private keys** in asymmetric cryptography
- Discuss the significance of **encryption key management**

Key Vocabulary

Term	Definition
Encryption	The process of converting plaintext into ciphertext to prevent unauthorised access
Decryption	The process of converting ciphertext back into plaintext
Symmetric cryptography	A cryptographic system where the same key is used for both encryption and decryption
Asymmetric cryptography	A cryptographic system where different keys are used for encryption (public) and decryption (private)
Public key	A key that can be shared with anyone; used for encryption
Private key	A key that must be kept secret; used for decryption

Term	Definition
Digital certificate	An electronic document that verifies the identity of an entity and binds a public key to that identity
Certificate Authority (CA)	A trusted organisation that issues digital certificates
Key management	The process of generating, storing, distributing, and revoking encryption keys
SSL/TLS	Security protocols that establish encrypted connections between web browsers and servers

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ATL Skills

Skill	Description
Communication	Explaining complex concepts clearly and concisely
Collaboration	Working effectively in groups to solve problems
Self-Management	Organising and managing learning independently
Research	Gathering and evaluating information from various sources
Thinking	Analysing and evaluating information, drawing conclusions

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2. What is Encryption?

Definition

> **Encryption is the process of converting plaintext (readable data) into ciphertext (unreadable data) to prevent unauthorised access. Decryption is the reverse process.**

Encryption Process

```text
┌──────────────────────────────────────────────┐
│              ENCRYPTION PROCESS              │
├──────────────────────────────────────────────┤
│                                              │
│   ┌────────────┐    ┌────────────┐    ┌────────────┐  │
│   │ Plaintext  │ ──▶ │ Encryption │ ──▶ │ Ciphertext │  │
│   │ (Readable) │    │   (Key)    │    │(Unreadable)│  │
│   └────────────┘    └────────────┘    └────────────┘  │
│                                              │
│               ───▶ (Transmission) ───▶            │
│                                              │
│   ┌────────────┐    ┌────────────┐    ┌────────────┐  │
│   │ Plaintext  │ ◀── │ Decryption │ ◀── │ Ciphertext │  │
│   │ (Readable) │    │   (Key)    │    │(Unreadable)│  │
│   └────────────┘    └────────────┘    └────────────┘  │
│                                              │
│   ✅ Confidentiality: Only authorised parties can read the data│
│                                              │
└──────────────────────────────────────────────┘
```

Why Encryption Matters

Purpose	Description
Confidentiality	Ensures only authorised parties can read the data
Integrity	Ensures data has not been altered during transmission
Authentication	Verifies the identity of the sender/receiver
Non-repudiation	Prevents a party from denying they sent a message

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3. Symmetric Cryptography

Definition

> Symmetric cryptography uses the same key for both encryption and decryption. Both sender and receiver must share the same secret key.

How It Works

```text
┌────────────────────────────────────────────────────────────┐
│                 SYMMETRIC CRYPTOGRAPHY                     │
├────────────────────────────────────────────────────────────┤
│                                                            │
│    Sender                      Receiver                    │
│                                                            │
│    ┌────────────┐              ┌────────────┐              │
│    │ Plaintext  │              │ Plaintext  │              │
│    └─────┬──────┘              └─────▲──────┘              │
│          │                           │                     │
│          ▼                           │                     │
│    ┌────────────┐  ◄────Same────►  ┌────────────┐          │
│    │  ENCRYPT   │       Key        │  DECRYPT   │          │
│    └─────┬──────┘                  └─────▲──────┘          │
│          │                               │                 │
│          ▼                               │                 │
│    ┌────────────┐  ───► (Transmission) ───► ┌────────────┐ │
│    │ Ciphertext │                           │ Ciphertext │ │
│    └────────────┘                           └────────────┘ │
│                                                            │
│    ✅ Fast and efficient                                   │
│    ❌ Requires secure key exchange                         │
│                                                            │
└────────────────────────────────────────────────────────────┘
```

Examples of Symmetric Encryption

Algorithm	Description
AES (Advanced Encryption Standard)	Widely used; strong; used for secure data storage
DES (Data Encryption Standard)	Older; less secure; largely replaced by AES

Algorithm	Description
3DES (Triple DES)	Improved version of DES; still used in some applications
Blowfish	Fast block cipher; used in password managers

Characteristics

Aspect	Description
Keys	One key used for both encryption and decryption
Speed	Fast — suitable for large amounts of data
Key Distribution	Challenge — how to securely share the key
Scalability	Poor — each pair of users needs a unique key (n(n-1)/2 keys for n users)
Best For	Encrypting large volumes of data at rest

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4. Asymmetric Cryptography

Definition

> **Asymmetric cryptography uses different keys for encryption and decryption: a public key (shared) and a private key (kept secret).**

How It Works

```text
┌─────────────────────────────────────────────────────────────┐
│                   ASYMMETRIC CRYPTOGRAPHY                  │
├─────────────────────────────────────────────────────────────┤
│                                                             │
```

```
│    Sender                     Receiver              │
│                                                     │
│   ┌───────────┐             ┌───────────┐          │
│   │ Plaintext │             │ Plaintext │          │
│   └─────┬─────┘             └─────▲─────┘          │
│         │                         │                 │
│         ▼                         │                 │
│   ┌───────────┐  ───────▶  ┌───────────┐          │
│   │  ENCRYPT  │  Receiver's │  DECRYPT  │          │
│   │           │  PUBLIC KEY │           │          │
│   └─────┬─────┘             └─────▲─────┘          │
│         │                         │                 │
│         ▼                         │                 │
│   ┌───────────┐  ───▶ (Transmission) ───▶ ┌───────────┐│
│   │ Ciphertext│                        │ Ciphertext ││
│   └───────────┘                        └───────────┘│
│                                                     │
│   ✅ No need to share secret keys                   │
│   ✅ Provides authentication and non-repudiation    │
│   ❌ Slower than symmetric encryption               │
│                                                     │
└─────────────────────────────────────────────────────┘
```

Public and Private Keys

Key	Purpose	Who Has It
Public Key	Encrypts messages	Anyone — can be shared freely
Private Key	Decrypts messages	Only the owner — must be kept secret

Examples of Asymmetric Encryption

Algorithm	Description
RSA	Most widely used; based on factoring large numbers

Algorithm	Description
ECC (Elliptic Curve Cryptography)	More efficient; used in mobile devices
Diffie-Hellman	Key exchange protocol

Characteristics

Aspect	Description
Keys	Two keys (public + private)
Speed	Slow — suitable for small amounts of data
Key Distribution	Easy — public keys can be shared freely
Scalability	Good — each user needs only one key pair
Best For	Secure key exchange, digital signatures, authentication

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5. Symmetric vs. Asymmetric Cryptography

Feature	Symmetric	Asymmetric
Keys	One key	Two keys (public + private)
Speed	Fast	Slow
Key Distribution	Difficult (must be secret)	Easy (public keys can be shared)
Scalability	Poor (n keys for n users)	Good (2n keys for n users)
Authentication	No	Yes
Non-repudiation	No	Yes
Best For	Large data encryption	Secure key exchange, digital signatures

Feature	Symmetric	Asymmetric
Examples	AES, DES, 3DES	RSA, ECC

Hybrid Approach

> Modern systems often use a hybrid approach: asymmetric cryptography is used to securely exchange a symmetric key, which is then used to encrypt the actual data.

```text
┌─────────────────────────────────────────────────────────────────┐
│                   HYBRID ENCRYPTION                             │
├─────────────────────────────────────────────────────────────────┤
│                                                                 │
│   Step 1: Asymmetric encryption is used to exchange a           │
│           symmetric key securely.                               │
│                                                                 │
│   Step 2: Symmetric encryption is used to encrypt the           │
│           actual data (fast).                                   │
│                                                                 │
│   ┌─────────────────────────────────────────────────────────┐   │
│   │  Example: HTTPS/TLS                                     │   │
│   │  - Asymmetric keys exchange the session key             │   │
│   │  - Symmetric keys encrypt the web traffic               │   │
│   └─────────────────────────────────────────────────────────┘   │
│                                                                 │
│   ✅ Combines speed of symmetric with security of asymmetric │
│                                                                 │
└─────────────────────────────────────────────────────────────────┘
```

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6. Digital Certificates

What is a Digital Certificate?

> A digital certificate is an electronic document that verifies the identity of an entity and binds a public key to that identity.

Key Components of a Digital Certificate

Component	Description
Subject	The entity the certificate is issued to (e.g., a website domain)
Public Key	The public key associated with the subject
Issuer	The Certificate Authority (CA) that issued the certificate
Validity Period	When the certificate is valid (start and end dates)
Serial Number	A unique identifier for the certificate
Digital Signature	The CA's signature (verifies authenticity)

How Digital Certificates Work

```text
┌─────────────────────────────────────────────────────────────────┐
│                DIGITAL CERTIFICATE PROCESS                      │
├─────────────────────────────────────────────────────────────────┤
│                                                                 │
│   Website: "I am google.com"                                    │
│              │                                                  │
│              ▼                                                  │
│   ┌─────────────────────────────────────────────────────────┐   │
│   │  Certificate Authority (CA)                            │   │
│   │  Verifies the website's identity                        │   │
│   │  Issues a digital certificate                           │   │
│   └─────────────────────────────────────────────────────────┘   │
│              │                                                  │
│              ▼                                                  │
│   ┌─────────────────────────────────────────────────────────┐   │
│   │  Digital Certificate                                    │   │
│   │  • Domain: google.com                                 │   │
│   │  • Public Key: <key>                                  │   │
│   │  • Issued by: DigiCert                              │   │
│   │  • Expires: 2026-08-01                              │   │
│   │  • Digital Signature                                  │   │
│   └─────────────────────────────────────────────────────────┘   │
```

```
│              │                                                   │
│              ▼                                                   │
│   Browser receives certificate                                  │
│   Browser verifies the CA's signature                           │
│   Browser trusts the website if signature is valid              │
│                                                                  │
└──────────────────────────────────────────────────────────────────┘
```

Certificate Authorities (CAs)

Aspect	Description
Definition	A trusted organisation that issues digital certificates
Role	Verifies the identity of entities and signs certificates
Trust Model	Browsers are pre-configured with a list of trusted CAs
Examples	DigiCert, Let's Encrypt, GlobalSign, Comodo
Chain of Trust	Root CA → Intermediate CA → Server Certificate

Types of Digital Certificates

Type	Purpose
SSL/TLS Certificate	Secures websites (HTTPS)
Code Signing Certificate	Authenticates software code
Email Certificate	Secures email communication
Client Certificate	Authenticates individual users

Digital Certificates vs. Digital Signatures

Aspect	Digital Certificate	Digital Signature
Purpose	Verifies identity	Verifies integrity and authenticity
Key Used	Public key bound to identity	Private key used to sign
Verification	Browser checks CA signature	Recipient checks with public key

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7. Key Management

Why Key Management Matters

> **Key management is the process of generating, storing, distributing, and revoking encryption keys. The security of encrypted data depends entirely on the security of the keys.**

Key Management Principles

Principle	Description
Key Generation	Keys should be generated using strong random number generators
Key Storage	Keys must be stored securely (hardware security modules, encrypted storage)
Key Distribution	Keys must be exchanged securely
Key Use	Keys should be used for a limited time
Key Revocation	Keys must be revocable if compromised
Key Destruction	Keys must be destroyed securely when no longer needed

Key Management Challenges

Challenge	Description
Key Theft	If a key is stolen, data is compromised
Key Loss	If a key is lost, encrypted data cannot be recovered
Key Revocation	Revoking compromised keys is difficult
Key Distribution	Sharing keys securely is challenging
Key Storage	Storing many keys securely is difficult

Best Practices

Practice	Description
Use Strong Keys	Use sufficiently long keys (e.g., 2048-bit RSA)
Rotate Keys	Change keys periodically
Secure Storage	Use hardware security modules (HSMs)
Access Control	Limit who can access keys
Key Backup	Securely back up keys
Key Revocation Plan	Have a plan for revoking compromised keys

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8. Real-World Applications

Application 1: HTTPS (Secure Web Browsing)

Step	Description
1	Browser requests a secure connection to a website

Step	Description
2	Website sends its digital certificate
3	Browser verifies the certificate with the CA
4	Browser uses the public key to encrypt a session key
5	Session key is used for symmetric encryption of web traffic
6	All communication is encrypted

Application 2: Email Encryption (PGP/GPG)

Aspect	Description
Public Key	Shared publicly (e.g., keyserver)
Private Key	Kept secret by the user
Encryption	Sender uses recipient's public key to encrypt
Decryption	Recipient uses private key to decrypt
Authentication	Digital signatures verify sender identity

Application 3: VPNs

Aspect	Description
Authentication	Server uses digital certificate to authenticate
Key Exchange	Asymmetric cryptography exchanges session keys
Data Encryption	Symmetric cryptography encrypts all traffic

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9. Lesson Sequence

Lesson Plan

Phase	Time	Activity
Introduction	5 min	Icebreaker: "When have you used encryption?"
Interactive Lecture	20 min	Symmetric vs. asymmetric cryptography; digital certificates; key management
Quizlet Live	10 min	Gamified reinforcement of key terms and definitions
Hands-On Activity	15 min	Caesar cipher, substitution cipher, online encryption tools; examine digital certificates
Wrap-up	10 min	Review key concepts; assign homework

Hands-On Activities

Activity	Description
Caesar Cipher	Encrypt and decrypt messages by shifting letters
Simple Substitution Cipher	Use a pre-defined key to substitute letters
Online Encryption Tools	Explore tools demonstrating different encryption algorithms
Examine Digital Certificates	View digital certificates from various websites (click the padlock icon in browser)

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10. Assessment Activities

Assessment Type	Description
Class Participation	Observe engagement and contributions
Quizlet Live	Assess understanding of key terms
Hands-On Activity	Evaluate ability to encrypt/decrypt using ciphers
Exit Ticket	Gauge understanding and identify areas needing clarification

Sample Exit Ticket Questions:

1. What is the difference between symmetric and asymmetric cryptography?
2. What is a digital certificate and what is its purpose?
3. Which key can be shared publicly: public key or private key?
4. Why is key management important in encryption?

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11. Differentiation

Level	Strategy
Support	Provide simplified explanations, visual aids, and step-by-step instructions
Challenge	Encourage exploration of more complex encryption algorithms, real-world case studies, or ethical implications of encryption

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12. Resources

- **Presentation** (Slide Deck)
- **eBook** (A2.4.4 with key questions)
- **Quizlet** (A2.4.4 vocabulary flashcards)
- **Online Encryption Tools** (for hands-on activities)

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13. Summary: Key Takeaways

Concept	Key Point
Symmetric Cryptography	Same key for encryption and decryption (fast, key distribution challenge)
Asymmetric Cryptography	Public key for encryption, private key for decryption (slower, no key sharing)
Hybrid Approach	Asymmetric to exchange keys, symmetric to encrypt data (e.g., HTTPS)
Digital Certificate	Binds a public key to an identity; issued by a Certificate Authority
Public Key	Shared freely; used for encryption
Private Key	Kept secret; used for decryption
Key Management	Generation, storage, distribution, and revocation of keys

> "*Encryption and digital certificates are the foundation of online security—they ensure that our data remains confidential, authentic, and trustworthy in an increasingly connected world.*"